Get ready for AI and Data Science Professional

Student Setup Guide | About 30 minutes | Costs nothing | Do this before Session 1

Every tool in this course is free. You never need a paid account, an API key, a graphics card or software to install. You need a laptop (Windows or Mac) with a browser and a charger. Free services change their limits from time to time. Where this guide gives a limit it says "at the time of writing", which means early November 2026. Check the provider's page for the current terms.

#	What	Needed for	Done
1	A Google account	Session 1	
2	Open Google Colab once, run a cell and save a copy	Session 5 (a try before Session 1 helps)	
3	Know how the data file, data.zip, reaches a notebook (Google Drive link)	Sessions 5 to 23	
4	Open the Lakeside workbook in Google Sheets or Excel	Sessions 1 to 4	
5	A free GitHub account	Session 23	
6	Streamlit Community Cloud (optional, after Session 23)	After Session 23	

1. A Google account

Google Colab (where you run Python) and Google Sheets both use a free Google account. If you have a Gmail address, you have one. If not, make one at accounts.google.com and write the password somewhere safe. If you do not want a Google account, tell your tutor before Session 1. There is a route that runs Python on your own laptop.

2. Open Google Colab once

1. Go to colab.research.google.com and sign in with your Google account.
2. Choose New notebook.
3. In the first cell type: print("Hello from Colab")
4. Press the play button beside the cell. You should see Hello from Colab under the cell.
5. Choose File, then Save a copy in Drive. Your tutor's notebooks open the same way: in Google Drive, double-click the notebook file, or choose Open with, then Google Colaboratory.

Menu names can change a little when Google updates Colab. If a name is missing, look for the same words in the nearby menu or ask your tutor.

> **Two menu items you will use in every session**
>
> Runtime, then Run all: runs every cell from the top.
>
> Runtime, then Restart session and run all: starts from a clean session. If the notebook finishes with no red text, it works from the top.

3. The data file: how data.zip reaches a notebook

Your tutor keeps data.zip on Google Drive and shares it with the setting "Anyone with the link". Each Python lab notebook starts with a cell headed "Run this cell first". The notebook downloads data.zip from the tutor's link and unpacks it. This is what to do:

1. Open the notebook and look at the first code cell. Near the top is a line that starts DATA_LINK = followed by quotation marks.
2. If a link is already between the quotation marks, you have nothing to paste. Run the cell.
3. If the quotation marks are empty and your tutor gives you a link, paste the link between them, then run the cell. Do not unzip anything.

4. If your tutor gives you the file itself instead of a link, leave DATA_LINK empty and run the cell. It prints "Choose the file data.zip from your computer" and shows a Choose files button. Click it and pick data.zip. If your browser renamed the file to data (1).zip, pick that one. It still works.
5. Wait a few seconds. The cell prints "Data files ready:" with a list of the csv and db files.
6. Open the folder icon on the left of Colab to see the files. They sit in the temporary space of your Colab session. Your Drive does not hold them.

If the download fails, check that you pasted the whole link and that you are signed in to the internet. Ask your tutor to confirm that the link is shared with "Anyone with the link". Because the files live in the session, the cell runs again each time you start a new session, open a new notebook, or Colab disconnects. The next page says what to do then.

4. The Lakeside workbook in Google Sheets or Excel

Sessions 1 to 4 are spreadsheet work. Your tutor will give you lakeside_for_sheets.xlsx (it is also inside data.zip). To use it in Google Sheets, upload it to Google Drive, open it, and choose File, then Save as Google Sheets. To use Excel or LibreOffice Calc, open the file directly. Look at the tabs along the bottom. In Session 2 you describe each one. The task lists are in the Sheets Lab Sheets document you receive in Session 1.

5. A free GitHub account

GitHub is a website that keeps a copy of your project online. You need an account in Session 23, and it is easier to make it now. Go to github.com, choose Sign up, and use your email address. Pick a username you would be happy to show an employer, because your course project will sit under it. A public repository can be read by anyone, so never upload personal data, passwords or anything your employer has not cleared. GitHub offers free accounts at the time of writing. Check the current terms if you want a private repository.

6. Streamlit Community Cloud (optional, after Session 23)

In Session 23 you write a small app. Streamlit Community Cloud can put it on the internet from your GitHub repository. You sign in with your GitHub account at share.streamlit.io. At the time of writing it is free for public apps, and an app that nobody visits for a while goes to sleep (reports say about 12 hours). That suits learning and demonstrations. It is not a place for a service a business depends on. Check the current limits and terms before you rely on it. Publishing is not part of the 45-minute lab. You do it afterwards.

Other tools you will meet

- TensorFlow Playground (Session 20): a web page, no account.
- sentence-transformers (Sessions 21 and 22): the notebook downloads a small model inside Colab once (about 90 MB at the time of writing). It needs a stable connection that day.
- A free chat assistant (optional): it can explain an error or suggest code, and you check what it says. Never paste personal or confidential data into one.

When Colab disconnects

It happens to everyone. Free Colab sessions end after you leave them idle for a while, after a long time connected, when the browser tab is closed, or when the laptop sleeps. Google does not publish fixed limits for the free service. At the time of writing, community reports give an idle disconnect of roughly 90 minutes and a longest session of roughly 12 hours, and both can change. Check the Colab FAQ for the current position.

Lost when it disconnects	The values held in memory (your variables and tables) and the files in the Colab Files panel, including data.zip contents and any csv you wrote.
Kept	The notebook itself, if you saved a copy in Drive. Anything you downloaded to your laptop.

1. Do not panic and do not retype. Choose Connect, or reload the page.
2. Run the data cell again. It downloads data.zip from the tutor's link, or asks you to choose the file.
3. Choose Runtime, then Run all (or Restart session and run all). Cells run from the top and rebuild your variables.

4. If a cell shows NameError, such as name df is not defined, a cell above it has not run in this session. Run the cells from the top.
5. Before you leave a lab, download the files you made (for example orders_clean.csv) and choose File, then Save a copy in Drive. Do this when the lab ends, not later.

If something goes wrong

What you see	What to do
Colab will not open	Try another browser (Chrome or Edge), and turn off any extension that blocks pages.
A cell shows red text	Read the last line first. It says what went wrong. We practise this in Session 5.
The data cell finishes but later cells cannot find orders.csv	Run the first cell again. If it asks for a file, choose data.zip. An unzipped folder does not work.
ModuleNotFoundError	The library is not installed in this session. Run pip install and the library name in a cell, then run the cell again. Ask your tutor if it still fails.
The page is stuck on Connecting	Check your internet connection, then reload. Free capacity can run short at busy times. Try again in a few minutes.

Data and privacy

Use only the fictional course data. Never put real customer, patient or staff data into a notebook, a chat assistant or an app you share.

What to bring

A laptop with a browser and a charger. A notebook and pen. A question from your own work that you would like data to answer: you will use it in Session 1.

Your other documents

Your tutor gives you these in Session 1: the Sheets Lab Sheets (the task lists for Sessions 1 to 4 and the presentation preparation in Session 24), the Glossary (every term the course teaches, with the spreadsheet idea it matches), and the Course Projects Guide (the projects you build, how to present your course project, the project checklist for feedback, and what the certificate of completion means).